

UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA (UPSA)

DEGREE PROGRAMME

REGULAR SCHOOL



END OF SECOND SEMESTER EXAMINATIONS- 2017/2018 ACADEMIC YEAR

FACULTY OF ACCOUNTING & FINANCE

DEPARTMENT OF BANKING AND FINANCE

LEVEL 200

BCPC 208: QUANTITATIVE METHODS

Date: Friday, May 24, 2019

TIME ALLOWED: 2 HOURS, 45 MINUTES

INDEX NUMBER

INSTRUCTION(S):

ATTEMPT ANY FOUR (4) QUESTIONS

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Question 1

Botch Potato is a product of the Cycy and Association. Producers in the area are able to switch back and forth between potato and wheat production depending on market conditions. As a result the producers of Botch Potato contracted an economist who estimated the demand curve for the association's product using data from 28 supermarket as follows:

$$Q_d = 90 - 1.5P$$

In addition, the economist also estimated the supply function for the product as:

$$Q_s = 3.5P - 35$$

You are to use this information to

- Determine the equilibrium level of prices and quantity demand. **3 marks**
- Represent your answer graphically and determine the welfare of economic agents. **6 marks**
- Suppose the government imposes a tax of ₦4 on every laptop sold, determine the new equilibrium price and quantity. **4 marks**
- Show how the economic agents share the tax. **1 mark**
- What is the amount of government tax revenue? **1 mark**

Question 2

A man had 3 children Beatrice, Emmanuel and Felicia, whom he gave an amount of 10000 to be shared among them. At the start of the year 2018, the three siblings decided to invest their share in an interest bearing account. They were able to negotiate different interest rate for their investment based on their negotiating skills and the amount of money they want to invest. In particular, the eldest of the three siblings, Beatrice was able to negotiate an interest of 5% per annum and thus invested an amount ₦x. In addition, the second eldest, Emmanuel was also able to negotiate an interest of 10% per annum for his investment. In effect, he invested an amount ₦y at the bank. The last born, Felicia, was able to negotiate an interest of 8% on her investment and thus invested an amount of ₦z at the bank. At the end of the year, 2018, the three siblings were able to earn a total interest of ₦800. In addition, supposed the amount of invested given to Felicia is 400 more than the amount invested by her two siblings combined.

- Write down the three linear equations. **2 marks**
- represent them in the matrix form $AX = B$ **1 mark**
- Find the determinant of the matrix A **3 marks**
- Find the amount of money given to each of the 3 siblings using Cramer's rule or otherwise. **9 marks**

Question 3

CyzyBotch Bridals produces bridal fans and wedding bouquets for two retail shops. In order to deliver a bridal fan requires 2 hours of production time, 3 hours for packaging and transportation requires three times the hours required for packaging. Also, to deliver a bouquet to the shop requires 4 hours of transportation time, half the time required for transportation is needed for packaging and five times the time needed for packaging is needed for producing a bouquet. To produce these two goods, the company incurs a cost of ₵45 and ~~₵90~~ ^{₵40} in order to produce a bridal fan and a bouquet respectively. In addition, the market conditions dictate the per unit price of bridal fans and bouquets at ₵135 and ₵165 respectively. Typically in a month, the company allocates at most 200 hours for production, 66 hours for packaging and 180 hours for transportation.

- Formulate a linear programming model for the above managerial problem. **3 marks**
- Find how many bridal fans and bouquets the company should produce in order to maximize profit. **8 marks**
- Determine the scarce resource(s). **1 mark**
- Find the shadow price of each of the scarce resources available to the firm. **3 marks**

Question 4

Stonewall and Associates is a monopolistic producer of two goods, The Botch and Mr. Minlah. In order to make informed decisions, the Chief Executive Officer of the company hired an economist to estimate the demand curves for the two products as follows:

$$P_1 = 50 - Q_1$$

$$P_2 = 95 - 3Q_2$$

Where Q_1 and Q_2 represents the quantities of The Botch and Mr. Minlah respectively and P_1 and P_2 represents the respective prices. In addition, the economist also estimated the joint total cost function for the firm as:

$$TC = Q_1^2 + 3Q_1Q_2 + Q_2^2$$

- Find the profit function for the firm. **2 marks**
- Find the values of Q_1 and Q_2 that maximizes the firms' profit by using the first and second order conditions. **10 marks**
- Find the price that the firm is supposed to charge for each good. **2 marks**
- What will be the maximum profit of the firm at the profit maximizing level of outputs? **1 mark**

Question 5

The University of Professional Studies, Accra (UPSA) was founded in 1965 as a private professional business education tuition provider. In 1999, by means of the Institute of Professional Studies Act, (Act 566), it became a tertiary institution with the mandate to provide tertiary and professional education in Accountancy, Management and related disciplines. Currently, the school has a student population of approximately 12,000. As part of its expansion program, the school now have a state-of-the-art Students' Center where food, snacks and drinks are served to students and staff. In effect, a sizeable number of UPSA students patronize Auntie Ama's Food Joint, which is located at the ground floor of the Students' Centre and produces plain rice and jollof rice. Suppose Auntie has a joint total cost function for plain rice and jollof rice as:

$$TC = 6Q_1 + 2Q_1Q_2 + 10Q_2$$

Where Q_1 and Q_2 denote plates of plain rice and jollof rice respectively. If P_1 and P_2 denote the corresponding prices then the demand equations are

$$P_1 = 40 - Q_1 + 2Q_2$$

$$P_2 = 20 + 4Q_1 - Q_2$$

- Find the profit function for Auntie Ama's Food Joint. **2 marks**
- Find the maximum profit if the firm is contracted to produce a total of 18 plates of either type. **9 marks**
- Find the price that the firm is supposed to charge for each good. **2 marks**
- Estimate the new optimal profit if the production quota increases to 22 plates. **2 marks**

Question 6

A manufacturing company has three processing lines. Each processing line combines various raw materials to produce the same output, Y. One of the processing lines can combine units of input (X) to produce units of output (Y) such that there is a linear relationship between X and Y. The Chief Executive Officer wants the output model to be estimated for this processing line. Given the values of X and Y (in thousands of units) in the table below, and as the Director of Data Management, you are to:

X	5	7	2	8	4	3	9	2	5	3
Y	8	12	6	14	8	9	18	5	10	7

- Use the least-squares method to compute the regression coefficients b_0 and b_1 . (1 mark)
- Write down the estimated equation and interpret the parameter estimates. (1 mark)
- Determine the value of the extent of relationship between input (X) and output (Y), and interpret your result. (1 mark)
- Find the value of the coefficient of determination and interpret your result. (1 mark)
- What will be the output level if the input usage is 15000? (1 mark)
- What percentage of the changes in output is due to other raw materials apart from X? (1 mark)
- Test at 1 percent significance level, whether the correlation between X and Y exist. (5 marks)

[Note: The table is 4 marks]

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$$b_1 = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}$$

$$b_1 = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$b_1 = \frac{\sum XY - \bar{X}\bar{Y}}{\sum X^2 - (\bar{X})^2}$$

$$b_0 = \bar{Y} - b_1\bar{X}$$

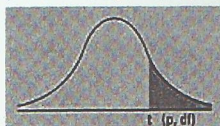
$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}}$$

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{\sum XY - \bar{X}\bar{Y}}{\sqrt{[\sum X^2 - (\bar{X})^2][\sum Y^2 - (\bar{Y})^2]}}$$

$$t^{cal} = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

Numbers in each row of the table are values on a t -distribution with (df) degrees of freedom for selected right-tail (greater-than) probabilities (p).



df/p	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0005
1	0.324920	1.000000	3.077684	6.313752	12.70620	31.82052	63.65674	638.6192
2	0.288675	0.816497	1.885618	2.919986	4.30265	6.96456	9.92484	31.5991
3	0.276671	0.764892	1.637744	2.353363	3.18245	4.54070	5.84091	12.9240
4	0.270722	0.740697	1.533206	2.131847	2.77645	3.74695	4.60409	8.6103
5	0.267181	0.726687	1.475884	2.015048	2.57058	3.36493	4.03214	6.8688
6	0.264835	0.717558	1.439756	1.943180	2.44691	3.14267	3.70743	5.9588
7	0.263167	0.711142	1.414924	1.894579	2.36462	2.99795	3.49948	5.4079
8	0.261921	0.706387	1.396815	1.859548	2.30600	2.89646	3.35539	5.0413
9	0.260955	0.702722	1.383029	1.833113	2.26216	2.82144	3.24984	4.7809
10	0.260185	0.699812	1.372184	1.812461	2.22814	2.76377	3.16927	4.5869
11	0.259556	0.697445	1.363430	1.795885	2.20099	2.71808	3.10581	4.4370
12	0.259033	0.695483	1.356217	1.782288	2.17881	2.68100	3.05454	4.3178
13	0.258591	0.693829	1.350171	1.770933	2.16037	2.65031	3.01228	4.2208
14	0.258213	0.692417	1.345030	1.761310	2.14479	2.62449	2.97684	4.1405
15	0.257885	0.691197	1.340606	1.753050	2.13145	2.60248	2.94671	4.0728
16	0.257599	0.690132	1.336757	1.745884	2.11991	2.58349	2.92078	4.0150
17	0.257347	0.689195	1.333379	1.739607	2.10982	2.56693	2.89823	3.9651
18	0.257123	0.688364	1.330391	1.734064	2.10092	2.55238	2.87844	3.9216
19	0.256923	0.687621	1.327728	1.729133	2.09302	2.53948	2.86093	3.8834
20	0.256743	0.686954	1.325341	1.724718	2.08596	2.52798	2.84534	3.8495
21	0.256580	0.686352	1.323188	1.720743	2.07961	2.51765	2.83136	3.8193
22	0.256432	0.685805	1.321237	1.717144	2.07387	2.50832	2.81876	3.7921
23	0.256297	0.685306	1.319460	1.713872	2.06866	2.49987	2.80734	3.7676
24	0.256173	0.684850	1.317836	1.710882	2.06390	2.49216	2.79694	3.7454
25	0.256060	0.684430	1.316345	1.708141	2.05954	2.48511	2.78744	3.7251
26	0.255955	0.684043	1.314972	1.705618	2.05553	2.47863	2.77871	3.7066
27	0.255858	0.683685	1.313703	1.703288	2.05183	2.47266	2.77068	3.6896
28	0.255768	0.683353	1.312527	1.701131	2.04841	2.46714	2.76326	3.6739
29	0.255684	0.683044	1.311434	1.699127	2.04523	2.46202	2.75639	3.6594
30	0.255605	0.682756	1.310415	1.697261	2.04227	2.45726	2.75000	3.6460
z	0.253347	0.674490	1.281552	1.644854	1.95996	2.32635	2.57583	3.2905
CI	———	———	80%	90%	95%	98%	99%	99.9%